CSE STUDY HUB

Data Structures & Algorithms

UNIT 2 · Stacks and queues

What this unit is about

This unit builds understanding of **Stacks and queues**. These notes are designed for quick learning and exam revision: learn the meaning, understand the working, practise an example, and finish with comparisons and key terms.

Syllabus coverage checklist

1. Stack ADT: LIFO principle, push, pop, peek and overflow/underflow

2. Stack implementation using arrays and linked lists

3. Infix, postfix and prefix expressions; precedence and associativity

4. Expression conversion and postfix evaluation using a stack

5. Queue ADT: FIFO principle, enqueue and dequeue

6. Linear queue, circular queue, deque and priority queue; applications

How to study: Read each topic once for meaning, underline definitions and formulas, reproduce diagrams/steps without looking, then answer the revision questions on Page 5.

Exam answer pattern

For a 5-mark answer: (1) definition, (2) labelled diagram or formula, (3) working/components in steps, (4) example/application, and (5) one advantage, limitation or comparison.

UNIT 2 NOTES · PART 1

Stack ADT: LIFO principle, push, pop, peek and overflow/underflow

A stack follows LIFO (last in, first out). push inserts at the top, pop removes the top, and peek reads it. Check overflow for fixed arrays and underflow before popping an empty stack.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Stack implementation using arrays and linked lists

A stack follows LIFO (last in, first out). push inserts at the top, pop removes the top, and peek reads it. Check overflow for fixed arrays and underflow before popping an empty stack.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Stacks and queues, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 2 NOTES · PART 2

Infix, postfix and prefix expressions; precedence and associativity

Infix, postfix and prefix expressions; precedence and associativity is a syllabus concept in Data Structures & Algorithms. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Expression conversion and postfix evaluation using a stack

A stack follows LIFO (last in, first out). push inserts at the top, pop removes the top, and peek reads it. Check overflow for fixed arrays and underflow before popping an empty stack.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Stacks and queues, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 2 NOTES · PART 3

Queue ADT: FIFO principle, enqueue and dequeue

A queue follows FIFO (first in, first out). Enqueue inserts at the rear and dequeue removes from the front. Circular queues reuse freed array positions; deques allow operations at both ends.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Linear queue, circular queue, deque and priority queue; applications

A queue follows FIFO (first in, first out). Enqueue inserts at the rear and dequeue removes from the front. Circular queues reuse freed array positions; deques allow operations at both ends.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Stacks and queues, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

QUICK REVISION & EXAM PRACTICE

Key takeaways

• **Stack ADT: LIFO principle, push, pop, peek and overflow/underflow** — know the definition, core working and one example.

• **Stack implementation using arrays and linked lists** — know the definition, core working and one example.

• **Infix, postfix and prefix expressions; precedence and associativity** — know the definition, core working and one example.

• **Expression conversion and postfix evaluation using a stack** — know the definition, core working and one example.

• **Queue ADT: FIFO principle, enqueue and dequeue** — know the definition, core working and one example.

• **Linear queue, circular queue, deque and priority queue; applications** — know the definition, core working and one example.

Worked-answer framework

Scenario: A question asks you to explain or apply a concept from Stacks and queues. **Step 1:** identify the relevant term and define it. **Step 2:** write the formula, components or ordered process. **Step 3:** apply it to the given data/scenario. **Step 4:** state the result and justify it. **Step 5:** mention one limitation or precaution.

Self-test questions

Q1. Define Stack ADT: LIFO principle, push, pop, peek and overflow/underflow and explain its importance.

Q2. Explain the working/steps involved in Stack implementation using arrays and linked lists.

Q3. Differentiate between two related concepts from Stacks and queues.

Q4. Give one practical application of Linear queue, circular queue, deque and priority queue; applications and discuss one limitation.

Q5. Write a structured 5-mark answer covering the main ideas of Stacks and queues.

Final checklist: I can define every topic · explain the process without notes · reproduce formulas/diagrams · solve one example · compare related concepts.